

A 28nm 32Kb SRAM Computing-in-Memory Macro With Hierarchical Capacity Attenuator and Input Sparsity-Optimized ADC for 4b Mac Operation

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Abstract—Computing-in-memory (CIM) is an emerging approach for alleviating the Von-Neumann bottleneck of latency and energy overheads, and improving energy efficiency and throughput. In this brief, we present a novel CIM macro aimed at improving the energy efficiency and throughput of edge devices when running 4b multiply-and-accumulate (MAC) operations. The proposed architecture uses (1) a customized 9T1C bit-cell in charge-domain computation for sensing margin improvement and compact design; (2) a hierarchical capacity attenuator for 4b weight accumulation without complicated controlling switches and signals for throughput improvement; (3) an input sparsity-sensing-based flash analog-to-digital converters readout scheme to improve energy efficiency and throughput. Fabricated in 28nm CMOS technology, the proposed 32Kb SRAM CIM macro demonstrates an average energy efficiency of 646.6 TOPS/W (normalized to 4b/4b input/weight) and a throughput of 1638.4 GOPS while achieving 84.89% classification accuracy on the CIFAR-10 dataset at 4b precision in inputs and weights.

Index Terms—SRAM, computing-in-memory (CIM), capacitive coupling, hierarchical capacity attenuator, input sparsity.

I. INTRODUCTION

WITH the Deep Convolutional Neural Networks (DCNN) model achieving unprecedented growth in scale [1], modern Machine Learning accelerators suffer from Von Neumann bottleneck due to a massive amount of data

movement from off-chip memory to on-chip processors [2]. Multiply-and-accumulate (MAC) with high regularity and parallelism, is the most common operation in DCNNs and dominates the computing power and latency. Computation-in-memory (CIM) overcomes the Von Neumann bottleneck and minimizes the communication energy and latency via performing computation inside the memory before reading out the stored data, which makes CIM increasingly appealing to DCNN acceleration.

Previous SRAM CIM macros [3], [4], [5], [6], [7] achieve the MAC operation by transistor currents. However, the computation based on current suffered from poor nonlinearity and variation, leading to a significant deterioration of DCNNs accuracies. Capacitor-based CIM SRAM [8], [9], [10], [11], [12], [13], [14], [15] has recently become popular because of its high energy efficiency for DCNNs and robustness against PVT variations. However, several challenges are remaining in capacitor-based CIM. Firstly, the design of bit-cell for dot-product requires tradeoffs between bit-cell area and computing dynamic range. References [8], [13] have 8T1C bit-cell suffer threshold loss in dynamic range, while [9], [11], [12] achieve a rail-to-rail dynamic range but their bit-cells are large. Secondly, capacitor-based CIM implements multi-bit weight by a charge sharing circuit in analog domain with switches array and extra controlling signal [7], [14], or in digital domain, using shifter and adder groups [8]. The former method has long sharing latency and complicated control, while the latter is heavy in power and area. Thirdly, the multi-bit analog-to-digital converter (ADC) for CIM is a trade-off between speed (throughput) and power (energy efficiency) consumption. Successive approximation-register (SAR) ADC [4], [9], [14] consumes less power with a low speed, while flash-ADC [5], [7], [8] consumes more power with a high speed. Noticing the high proportion of low MAC values in DCNNs, [6] and [10] proposed an algorithm-adaptive low MAC-aware ADC scheme and a prioritized-hybrid ADC to improve energy efficiency, respectively. They are based on SAR and single-slope ADC.

To mitigate these challenges, in this brief, we present a fully-capacitive-coupling SRAM CIM macro with the following features: (1) a new 9T1C bit-cell for dot-product with fewer transistors and rail-to-rail dynamic range; (2) a hierarchical capacity attenuator (HCA) structure for 4b-weight accumulation without additional switches and complicated

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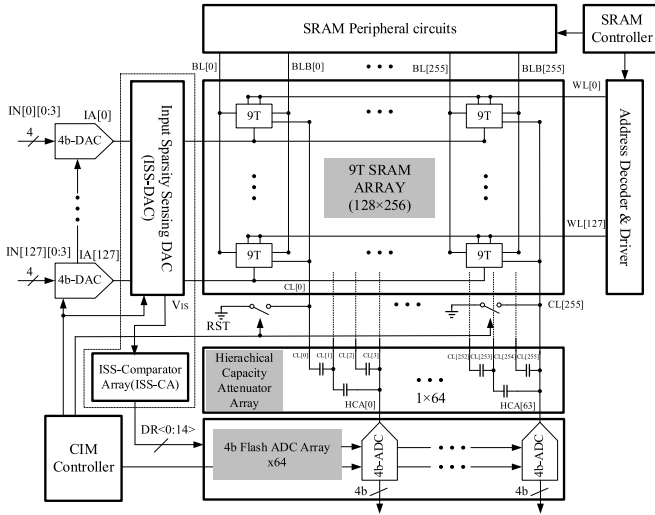


Fig. 1. Proposed SRAM-CIM macro architecture.

controls; (3) an input sparsity sensing (ISS) based flash-ADC readout scheme to minimize the expensive AD conversions. This brief is an extended version of [15], providing further detailed circuit descriptions, measurement results after fabrication, and more discussion.

II. PROPOSED SRAM CIM MACRO

A. Overall Architecture

Fig. 1 shows the proposed SRAM CIM macro architecture, which includes a 128×256 9T1C bit-cell array, SRAM read/write peripherals, an address decoder driver, 128×1 4b input DACs, an input sparsity sensing DAC (ISS-DAC), a 15×1 input-sparsity sensing comparator array (ISS-CA), a 1×64 HCA array, a 1×64 4b Flash ADC array, a 1×64 reset footers and a CIM/SRAM controller. The macro performs fully parallel MAC operations, with all rows asserting 4b inputs, computing vector-matrix multiplication of 128 inputs and 64 128×4b weights per cycle. ISS-based Flash ADCs convert the analog results to 4b digital values.

B. Proposed 9T1C Bit-Cell

Fig. 2 shows a column of 9T1C bit-cells, a table of AND operands, a table of 4b input for analog voltage input, and the computation waveform. The 9T1C bit-cell includes a traditional 6T SRAM bit-cell, a transmission gate (TG), an n-type MOS transistor (Mn), and a cap C. The cap is ~1.33f MOM cap placed above the bit-cell without area overheads. 1b weight is stored in the bit-cell and 4b input is applied as an analog voltage via 4b DAC. Like the 10T1C bit-cell [9], [11], the proposed 9T1C bit-cell has a rail-to-rail dynamic range, which is larger than the 8T1C bit-cell [8], [13].

As shown in Fig. 3(a), the computing includes two phases: reset and evaluation. Single-cell operation begins with the reset to GND for cap C. If Q=0, the Mn connected with QB opens and pulls the node Mult[i] down to GND. If Q=1, TG opens and pulls the node Mult[i] down to IA, which is GND in the reset phase. After the reset, the analog voltage is applied to IA. If and only if Q=1 can the analog voltage be transferred to the node Mult[i]. Multiple cells operation in a column is parallelized to multiple rows. Multiple weights are stored in multiple cells on a column and multiple inputs are applied as

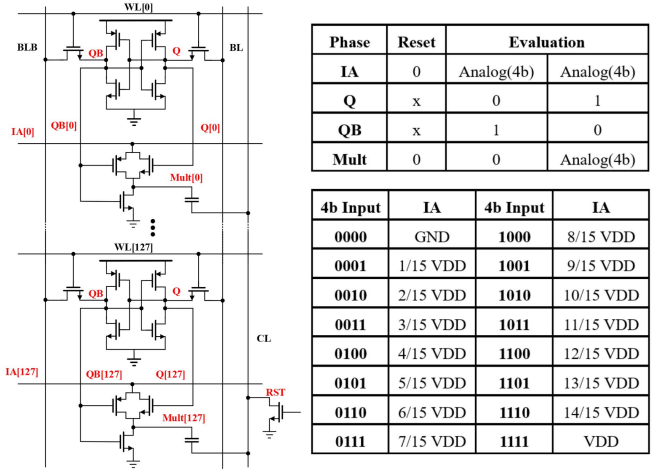


Fig. 2. Proposed 9T1C bit-cell, 4b input and 1b weight MAC table, and input analog voltage with 4b encoding table.

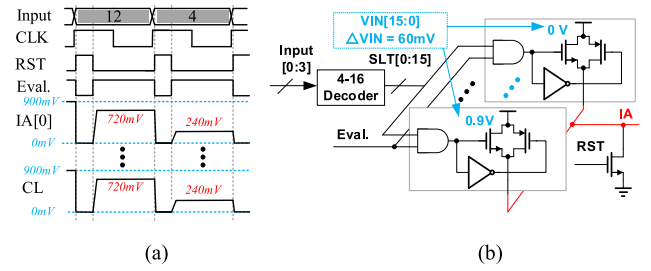


Fig. 3. (a) Single column operation waveform, and (b) Proposed 4b DAC.

driving voltage in IAs in the evaluation phase. Based on the principle of capacitor coupling, voltage V_{CL} is given by (1), where IA_i is the input of the i th row, w_i is the stored data in the i th row, and $IA_{max,i}$ is the biggest input, equaling 15 for 4b input. V_{CL} is proportional to the MAC result of 1b weights and 4b inputs.

$$V_{CL} = \frac{\sum_{i=0}^{127} IA_i w_i}{\sum_{i=0}^{127} IA_{max,i}} V_{DD} \quad (1)$$

Fig. 3(b) shows the digital-switch-based 4b DAC for 9T1C bit-cell, which includes a 4-16 decoder, 16-level voltages, and controlled switches array. During the evaluation phase, the decoder will select one of the DAC levels based on the input and apply it to IAs. With the DAC, 9T1C bit-cell, which can take analog inputs, are provided 4b input data with low LSB voltage in one cycle, which benefits the energy efficiency and throughputs.

C. Proposed Hierarchical Capacity Attenuator (HCA)

The MAC of 4b input and 4b weight is shown in Fig. 4, with each significant 1b weight multiplied by a 4b input and summed via the proposed HCA structure. The HCA includes two $\frac{1}{2}$ capacity attenuators (CA) and a $\frac{1}{4}$ CA. The $\frac{1}{2}$ CA sums the $W[0](W[2])$ and $W[1](W[3])$ weight pair with an attenuation coefficient(AC) of 0.5, while the $\frac{1}{4}$ CA sums the $W[0:1]$ and $W[2:3]$ weight pair with AC=0.25. The AC is determined by the proportion of the weight. With the AC of 0.5 and 0.25, the C_{w01}/C_{w23} and $C_{w01/23}$ are calculated to be 128C and 64C, respectively. C_{w01}/C_{w23} and $C_{w01/23}$ are integral multiples of C, which makes layout matching easier.

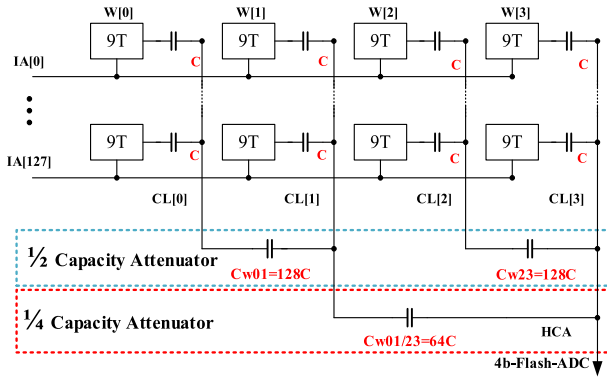


Fig. 4. Proposed HCA for 4b-input-4b-weight MAC implementation.

In reset phase, both the top and bottom plates of capacitors are reset to GND. In evaluation phase, all 4b inputs are fed in via IAs and clamped one plate of the bit-cell capacitor to a fixed voltage. When the coupled capacitor array achieves an equilibrium potential, the computation result V_{HCA} is generated.

The result is given by (2), where IA_i is the input of the i th row, $w_{i,j}$ is the stored data in the i th row and the j column, and $IA_{max,i}$ is the biggest input, equaling 15 for 4b input. V_{HCA} is proportional to the MAC result of 4b weights and 4b inputs.

$$V_{HCA} = \frac{\sum_{j=0}^3 \sum_{i=0}^{127} 2^j IA_i w_{i,j}}{\sum_{j=0}^3 \sum_{i=0}^{127} 2^j IA_{max,i}} V_{DD} \quad (2)$$

Compared to the previous charge-sharing method [9], [13], the proposed HCA has no more switch array for intermediate storage and charge sharing. Therefore, HCA has not only a simpler structure but also less computing time. Because HCA achieves an equilibrium potential in one step, while charge sharing method achieves via multi-step sharing with larger transistor resistors on the charge redistribution path.

D. Proposed ISS-Based Flash-ADC Readout Scheme

To acquire high throughput, a flash ADC is chosen for MAC 4b readout. However, flash ADC consumes more energy than other ADCs. Noticing that the whole dynamic range of MACs is seldom reached in the computation, especially when the input data is sparse, we proposed an ISS-based flash-ADC readout scheme based on real-time input sparsity characteristics to minimize the comparison energy. The scheme estimates the sum and quantifies the 128 4b inputs regardless of the stored weights, allowing it to skip redundant comparator work.

The proposed readout scheme is shown in Fig. 5(a). It consists of an ISS-DAC and an input sparsity 1×15 ISS-CA array. The ISS-DAC includes 128 capacitors, which are the same as in 9T1C bit-cell and have no area cost because of the high-level metal placement. The ISS-DAC converts 128 4b input into voltage V_{IS} , representing the input sparsity. The result is given by (3).

$$V_{IS} = \frac{\sum_{i=0}^{127} IA_i}{\sum_{i=0}^{127} IA_{max,i}} V_{DD} \quad (3)$$

Then V_{IS} compares with the same group of references used in 4b flash ADC by ISS-CA. A 15b thermometer code $DR[0:14]$ representing the quantized input sparsity is generated. The 15b code determines the working states of 15

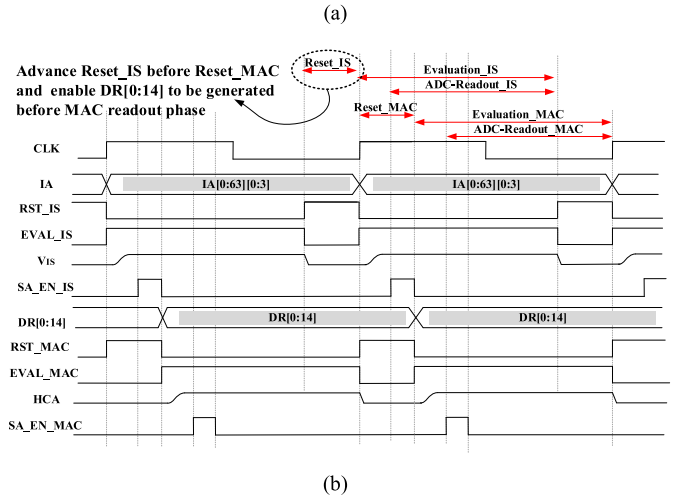
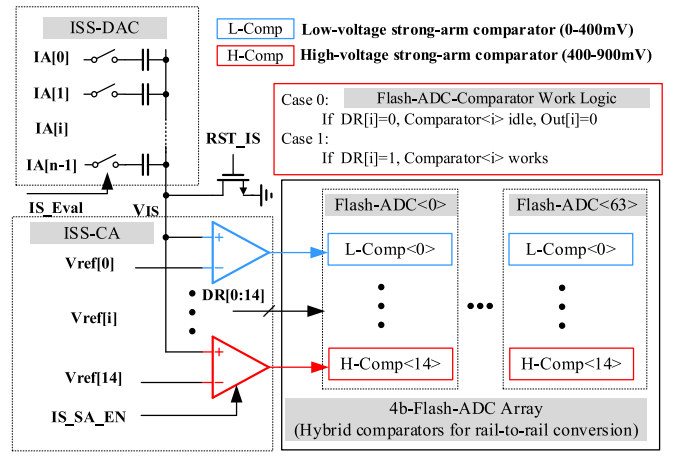


Fig. 5. (a) Proposed ISS-based 4b flash ADC readout scheme, and (b) Timing diagram of ISS scheme operation and MAC operation.

comparators in a flash ADC in readout phases. The control logic is shown in the upper box in Fig. 5(a). When the code is 0, the comparator will be skipped and the compared result is set to 0. When the code is 1, the comparator will work normally and generate the result. The proposed scheme skips redundant comparator work and saves energy at the area and energy cost of ISS-DAC and ISS-CA. Thus, energy efficiency gains only when the input is sparse to a certain extent. Note that the ISS-based readout scheme used in this brief will not slow down the throughput. We advance Reset_IS before Reset_MAC operation, enabling $DR[0:14]$ to be generated before MAC readout phase in Fig. 5(b).

III. MEASUREMENT RESULTS AND ANALYSIS

The prototype chip for the proposed SRAM CIM macro is implemented in a standard 28-nm CMOS process. As shown in Fig. 6, the capacity of the macro is 32Kb and the core area is 0.11018mm^2 . The bit-cell area is $0.994\text{ }\mu\text{m}^2$. Under a supply voltage of 0.9V, the operation frequency can reach 100MHz. Fig. 6 also gives the layout floorplan of the proposed 9T1C and HCA for 4b weight. For the symmetry and match of layout, firstly, C_{w01}/C_{w23} and C_{w02}/C_{w13} are cut into 128- and 64- bit-cell level capacitors, respectively. Secondly, we switch column W2 and column W3. Thirdly, we introduce the 64 dummy capacitors interleaved with partial C_{w02}/C_{w13} capacitors.

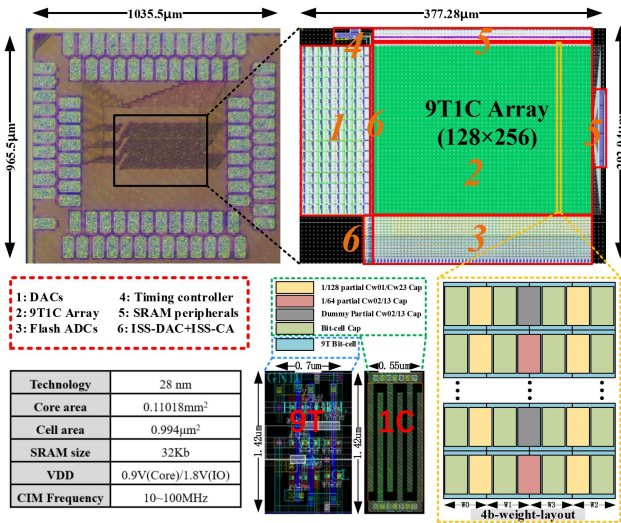


Fig. 6. Die photograph and layout of the fabricated SRAM CIM macro.

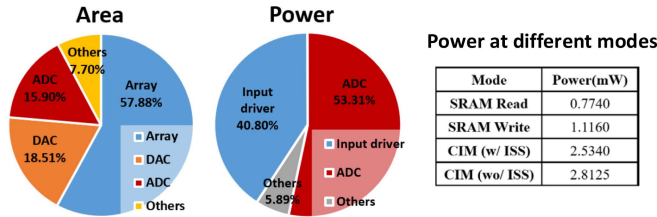


Fig. 7. Area and power breakdown (w/ISS) (others include control block, ISS-DAC, ISS-CA, and WL decoder), and power at different modes.

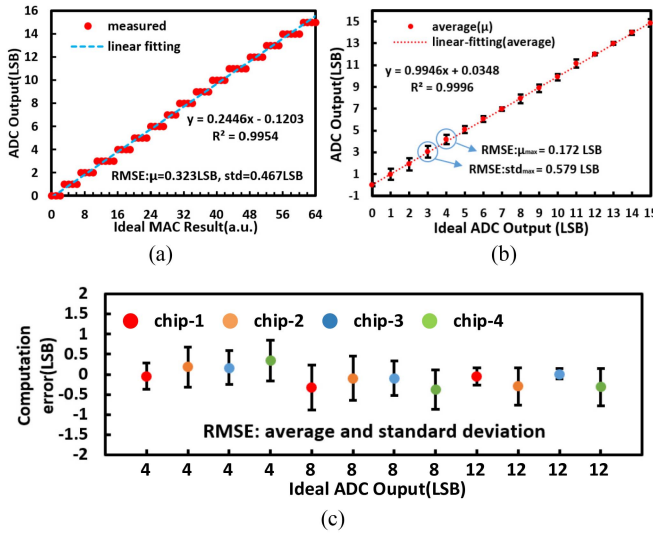


Fig. 8. (a) Measured intra-channel computation linearity and error (one channel), (b) measured computation intra-macro error (64 channels in one chip), and (c) measured computation macro-to-macro error across 4 chips. (One LSB is defined for the output resolution of the 4b ADC).

The layout floorplan is not the most compact implementation and can be further optimized. The area and power breakdown are shown in Fig. 7. The array area efficiency is 57.88%. DAC and ADC have almost the same area efficiency. 53.31% of power is consumed by ADCs, 40.8% by input driver (DAC and array charging/discharging), and 5.89% by others (w/ISS).

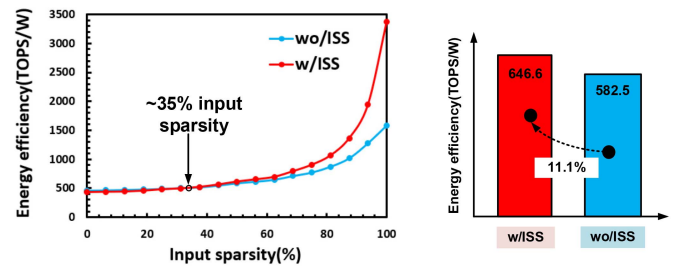


Fig. 9. Measured energy efficiency vs. input sparsity with an average energy efficiency of 646.6 (w/ISS) and 582.5(wo/ISS) TOPS/W.

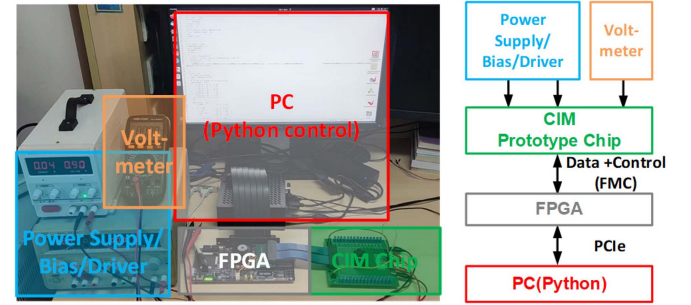


Fig. 10. FPGA-based test environment, and data flow diagram.

TABLE I
DATASET AND DCNN MODEL FOR ACCURACY MEASUREMENT

Dataset	CIFAR-10
VGG-like Neural Network topology	128C3-128C3-MP2-256C3-256C3-MP2-512C3-512C3-MP2-1024FC-10FC
Baseline accuracy(4b)	89.32%
Measurement accuracy	84.89%

In 100MHz CIM mode, the total power is 2.5340 mW and 2.8125 mW for w/ISS and wo/ISS scheme, respectively.

Fig. 8(a) shows the intra-channel computation linearity and error, which exhibits good linearity and a reasonably small root mean square error (RMSE) average (standard deviation) is 0.323 (0.467) LSB. Fig. 8(b) shows the computation intra-macro error among all 64 channels in a chip with sweeping MAC results. The maximum RMSE average (standard deviation) is 0.172 (0.579) LSB. We also measured the macro-to-macro computation error across 4 chips at three ideal ADC outputs (4, 8, and 12), as shown in Fig. 8(c). The result of three samples shows normal computation macro-to-macro error.

Fig. 9 shows the measured energy efficiency at different input sparsity. When at higher input sparsity (>35%), the result of w/ISS mode is significantly larger than wo/ISS mode, thanks to a large amount of skipped comparators in the ISS scheme. The average energy efficiency in w/ISS mode is 646.6 TOPS/W, which is 11.1% higher than that in wo/ISS mode due to fewer working comparators by the ISS scheme. The computing linearity and DCNN accuracy test are processed with the FPGA board and PC as shown in Fig. 10. The weights and input activations are loaded to FPGA together via the FMC port. The weights are written into the test chip in SRAM mode first. Then the input activations are given to the chip frame by frame, and the output activations are collected by FPGA and sent to the PC in succession through the PCIe Port.

TABLE II
COMPARISON TABLE WITH PRIOR SRAM-CIM WORKS

Previous work	JSSC'20 [7]	JSSC'20 [8]	VLSI'21 [9]	ISSCC'21 [10]	CICC'22 [13]	CICC'22 [14]	This work
Technology	7nm	65nm	28nm	28nm	65nm	28nm	28nm
Macro size	4kb	16kb	288kb	384kb	4kb	16kb	32kb
Bit-cell area(μm^2)	0.053	2.909	0.936	0.245 ¹	N/A	0.80275	0.994
Macro size(mm^2)	0.0032	0.081	0.51	N/A	0.2013	0.031	0.1101
Supply voltage(V)	0.65-1	1/0.8/0.6 ²	0.9/0.6	0.7-0.9	1.0/0.85	0.8/1.0	0.9
Computation mode	Current & Charge(share) ²	Charge (coupled)	Charge (coupled)	Charge (share)	Charge (coupled)	Charge (share)	Charge (coupled)
Cell structure	8T	8T1C	10T1C	6T+LCC	8T1C	9T1C	9T1C
Output dynamic range(V)	0.13~0.8V ³	0-0.8	0-0.9	0-(VDD-V _{thn}) ⁴	0~0.8	0~0.9	0-0.9
Input/Weight/Output bit width (b)	4/4/4	1/1/5	5/1/8	4/4/12	4/4/5	4/4/6	4/4/4
MAC parallelism	64	256	1152	16	64	128	128
Output channels	16	64	256	16	16	32	64
Throughput(GOPS) ^{5,6}	372.4	102.375	384	128	128	56	1638.4
Energy efficiency (TOPS/W) ^{5,6}	351	41.97	362.25	60.28-94.31	133.58	177	646.6
Measured Accuracy(CIFAR-10)	88.52%	85.5%	91.1%	N/A	97.98%(MNIST)	91.92%	84.89%⁸

¹Average area considered local computing cell, ²Current computation for 1b weight MAC and charge-sharing for multi-bit weight accumulation, ³Estimated from the description, ⁴Estimated from the proposed structure with NMOS as transmission gate, ⁵Normalized to 4b/4b input/weight operation, ⁶One MAC is counted as two operations (multiplication and addition), ⁷1V(array)/0.8V(driver)/0.6V(ADC), ⁸4b input precision and 4b weight precision.

A VGG-like DCNN model in [16] was applied to CIFAR-10 for training and inference. The measurement accuracy is 84.89% with an accuracy loss of 4.43%. Table I shows the comparison of recent SRAM CIM works. The proposed CIM macro achieves the highest energy efficiency and throughput in the table, which have improved by 1.785 $\times$ and 4.27 $\times$, respectively.

IV. CONCLUSION

In this brief, we present a 28 nm SRAM CIM macro aimed at improving the energy efficiency and throughput of edge devices for DCNNs running 4b MAC operations. To improve sensing margin, a customized 9T1C bit-cell in charge-domain computation is proposed. To improve energy efficiency and throughput, we proposed an HCA architecture for 4b weight accumulation without complicated control in charge sharing method. To minimize the expensive AD conversion and ensure high throughput, an input sparsity-sensing-based flash-ADC readout scheme is proposed, allowing to skip redundant comparators. The proposed design supports 8192 4b $\times$ 4b MAC computations simultaneously. Fabricated in 28nm CMOS technology, our 28 nm prototype chip operates up to 100 MHz at 1.2V. The measurement results show an average energy efficiency of 646.6 TOPS/W and a throughput of 1638.4 GOPS, which have improved by 1.785 $\times$ and 4.27 $\times$ compared with prior CIM works, respectively. The measured accuracy of the test chip on the CIFAR-10 test set is 84.89% at 4b input and 4b weight precision. Therefore, the proposed macro claims to be a promising CIM candidature for high throughput and energy-efficient AI edge devices.

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